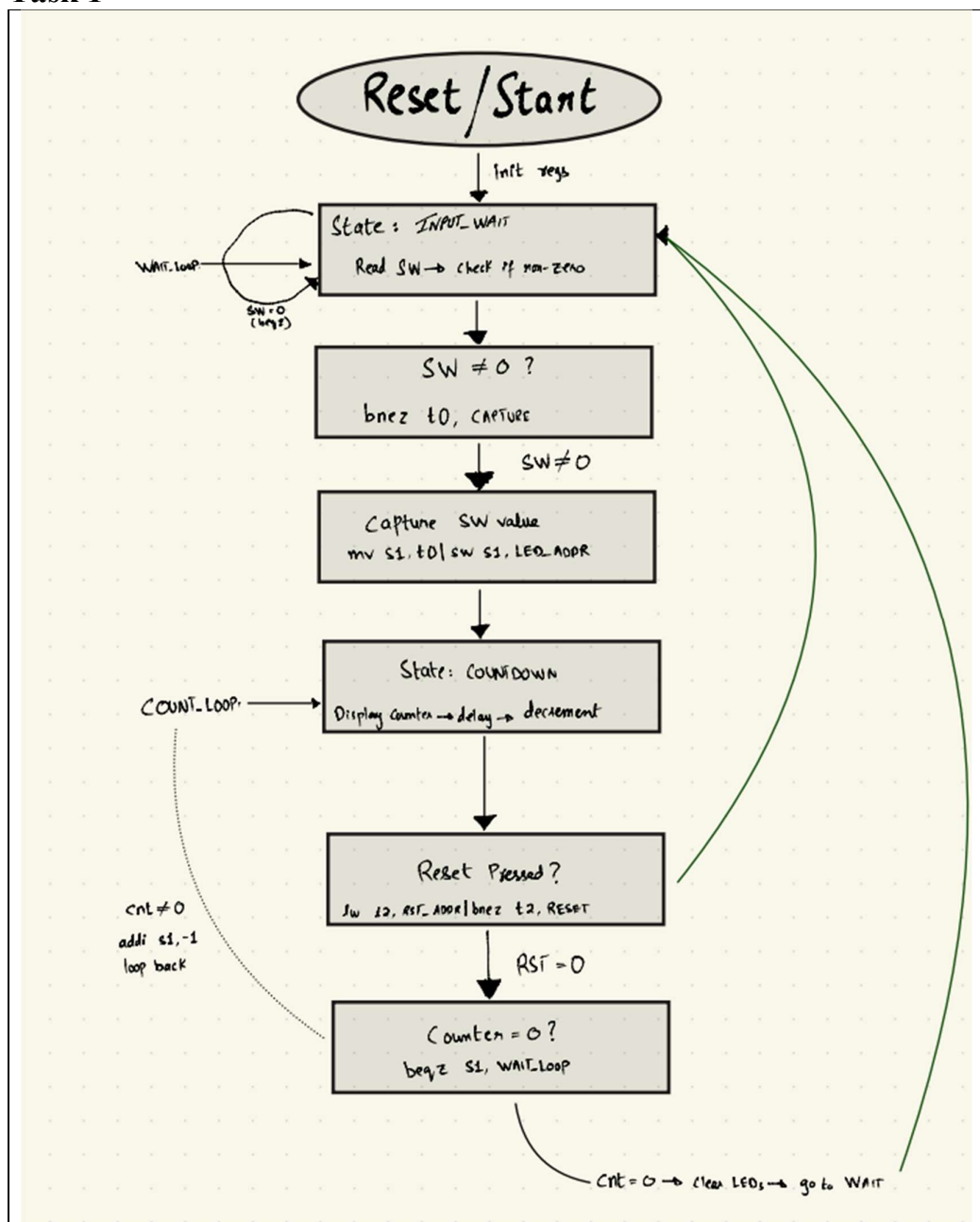




Lab 10 – Worksheet

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Task 1





Task 2

```
addi x28, x0, 5
addi x2, x0, 511
addi x5, x0, 768
addi x6, x0, 512
sw x28, 0(x5)
INPUT_WAIT:
sw x0, 0(x6)
POLL_SWITCH:
lw x11, 0(x5)
beq x11, x0, POLL_SWITCH
add x10, x11, x0
jal x1, COUNTDOWN
beq x0, x0, INPUT_WAIT
COUNTDOWN:
addi x2, x2, -8
sw x1, 4(x2)
sw x12, 0(x2)
add x12, x10, x0
COUNT_LOOP:
sw x12, 0(x6)
beq x12, x0, COUNT_DONE
addi x12, x12, -1
addi x13, x0, 3
DELAY:
addi x13, x13, -1
bne x13, x0, DELAY
beq x0, x0, COUNT_LOOP
COUNT_DONE:
sw x0, 0(x6)
lw x12, 0(x2)
lw x1, 4(x2)
addi x2, x2, 8
jalr x0, 0(x1)
```

Integer

x00 (zero) = 0x00000000

x01 (ra) = 0x00000000

x02 (sp) = 0x000001FF

x03 (gp) = 0x10000000

x04 (tp) = 0x00000000

x05 (t0) = 0x00000300

x06 (t1) = 0x00000200

a0 (x10)

0x00000005

a1 (x11)

0x00000005

a2 (x12)

0x00000000



a0 (x10)	0x00000005	a0 (x10)	0x00000005
a1 (x11)	0x00000005	a1 (x11)	0x00000005
a2 (x12)	0x00000005	a2 (x12)	0x00000004
a0 (x10)	0x00000005	a0 (x10)	0x00000005
a1 (x11)	0x00000005	a1 (x11)	0x00000005
a2 (x12)	0x00000003	a2 (x12)	0x00000002
a0 (x10)	0x00000005	a0 (x10)	0x00000005
a1 (x11)	0x00000005	a1 (x11)	0x00000005
a2 (x12)	0x00000001	a2 (x12)	0x00000000

decrementing and then staying stuck at input waiting state

Task 3

Required Verilog Files attached in the folder uploaded in Github

Synthesis proof:

Synthesis				
Synth Design (synth_design)				
synth_1_synth_report_utilization_0	report_utilization		3/24/26, 11:39 AM	6.9 KB
synth_1_synth_synthesis_report_0			3/24/26, 11:39 AM	13.2 KB



Assessment Rubric

Lab 09: Implementing Nested Procedures and Sorting

	Task No.	LR 2 Code	LR 5 Results	LR 11 (Design)
In-Lab	Pre-Lab	-	/15	-
	Task 1	-	-	/10
	Task 2	/20	/20	-
	Task 3	-	/15	-
Total Points		/20	/50	/10
CLO Mapped		CLO 1		

Affective Domain Rubric		Points	CLO Mapped
AR7	Report Submission & Git Upload	/10 & /10	CLO 1

CLO	Total Points	Points Obtained
1	100	
Total	100	

For description of different levels of the mapped rubrics, please refer to the Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics provided here.



Lab Evaluation Assessment Rubric

#	Assessment Elements	Level 1: Unsatisfactory	Level 2: Developing	Level 3: Good	Level 4: Exemplary
LR2	Program/Code/ Simulation Model/ Network Model	Program/code/simulation model/network model doesnot implement the requiredfunctionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulation model/network model has some errors and doesnot produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full commandon the basic tools of the software.
LR5	Results & Plots	Figures/ graphs / tables are not developed or are poorlyconstructed with erroneousresults. Titles, captions, units are not mentioned. Data is presented in anobscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are notaccurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minorerrors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captionsand proper units are mentioned. Data presentation is systematic.
AR9	Report Content/Code Comments	No summary provided. The number/amount of tasks completed below the level of satisfaction and/or submitted late	Couldn't provide good summary of in-lab tasks. Some major tasks were completed but not explained well. Submission on time. Some major plots andfigures provided	Good summary of In-lab tasks. All major tasks completed except few minor ones. The work is supported by some decent explanations, Submission on time, All necessary plots, and figures provided	Outstanding Summary of In-Lab tasks. All task completed and explained well, submitted on time, good presentation of plotsand figure with proper label, titles and captions